# Active Learning

## MSEN 489/655 - Materials Design Studio

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# Outline


1. Active Learning in Materials Design
2. Expected Improvement (EI)
3. Batch Active Learning: qExpectedImprovement
4. Conclusions

# Active Learning in Materials Design

# What is active learning?

## Core idea

Active learning is a sequential strategy where we do **not** choose experiments uniformly or randomly. Instead, we use the current model to decide **which experiment would be most valuable next**.

- Build a surrogate model from the data collected so far
- Quantify both **prediction** and **uncertainty**
- Choose the next point to:
  - **exploit**: test what already looks promising
  - **explore**: test where uncertainty is still high
- Update the model after each new experiment

## In one sentence

Active learning tries to maximize the **information gained per expensive experiment**.

# Why is active learning useful in materials design?

- Materials discovery is usually an **expensive black-box problem**

$$(\text{composition, processing}) \longrightarrow \text{property}$$

- Each query requires synthesis, heat treatment, characterization, and analysis
- Data are often: small, noisy, heterogeneous, expensive to acquire.
- So the question is not only *"what is the best material?"* but also *"which experiment should I run next?"*

**What active learning gives us**

- faster optimization
- better coverage of design space
- reduced experimental waste

**Typical use cases**

- alloy composition design
- process optimization
- adaptive characterization
- closed-loop experimentation

# Active learning loop

Model → Choose next experiment → Run experiment → Update data

- **Model:** a Gaussian process (GP) is a common surrogate
- **Choose:** an acquisition function decides where to sample next
- **Run:** perform the experiment or simulation
- **Update:** incorporate the new observation and repeat

**Key practical message**

This is especially valuable when the budget is small and every experiment matters.

# Expected Improvement (EI)

# Improvement

## Goal

Suppose we want to **maximize** an unknown objective $f(x)$, and the best observed value so far is $y^+$.

## Improvement at a candidate point $x$

$$I(x) = \max(f(x) - y^+, 0)$$

Since $f(x)$ is unknown, we use the GP posterior:

$$f(x) \mid \mathcal{D} \sim \mathcal{N}(\mu(x), \sigma^2(x))$$

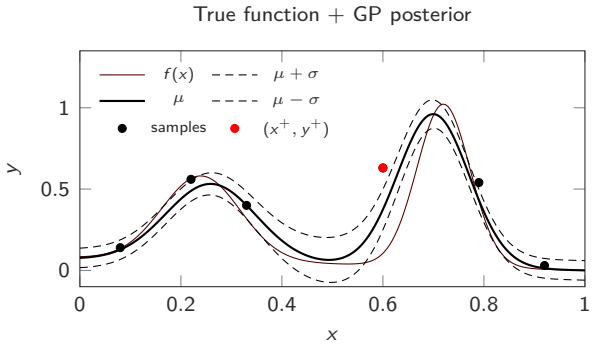
# Expected Improvement (EI)

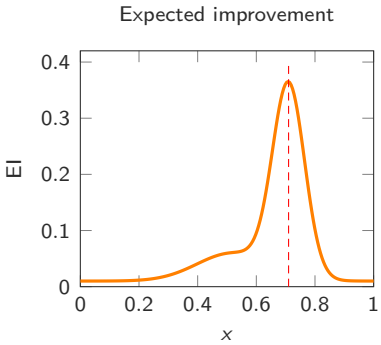
**Expected Improvement at a candidate point $x$**

$$\mathrm{EI}(x) = \mathbb{E}[I(x)] = (\mu(x) - y^+)\Phi(Z) + \sigma(x)\phi(Z), \qquad Z = \frac{\mu(x) - y^+}{\sigma(x)}$$

- $\mu(x)$ large $\Rightarrow$ **exploitation**
- $\sigma(x)$ large $\Rightarrow$ **exploration**
- EI balances both automatically

# EI example 1: dominated by a very good point (exploitation)

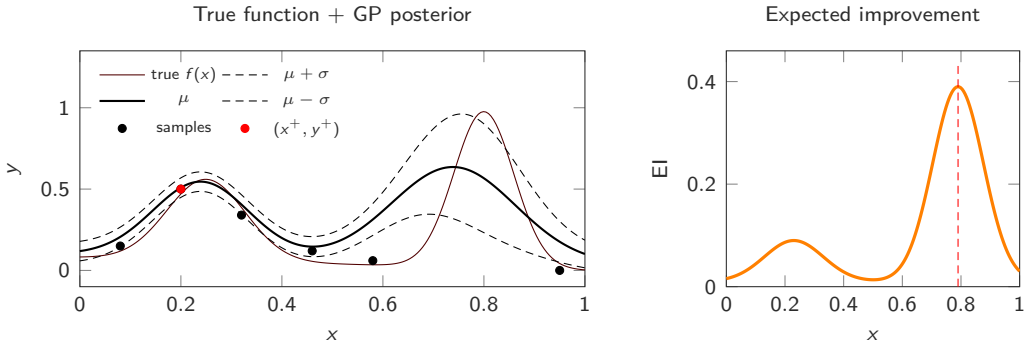



**Interpretation**

A point near $x \approx 0.7$ already looks **very good**. Even with only moderate uncertainty, EI is large because the GP mean is close to improving on the incumbent.

# EI example 2: dominated by high variance (exploration)




## Interpretation

GP mean is **not** the best near $x \approx 0.8$, but uncertainty is very large. EI becomes large because there is a real chance that this unexplored region contains a better material.

# EI takeaway

**Case 1: exploitation-dominated**

- High posterior mean
- Moderate uncertainty
- EI concentrates near the current promising region

**Case 2: exploration-dominated**

- Mean is not necessarily best
- Uncertainty is large
- EI prefers learning where the model is still unsure

**Main message**

The same acquisition rule can choose either "go where it already looks best" or "go where uncertainty could hide something better".

# Batch Active Learning: qExpectedImprovement

# From EI to qExpectedImprovement (qEI)

## Why do we need batch acquisition?

In many materials workflows we can run **multiple experiments in parallel**:

- several furnace conditions
- multiple compositions on the same synthesis batch

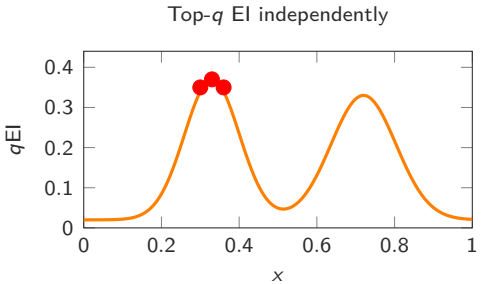
## Definition

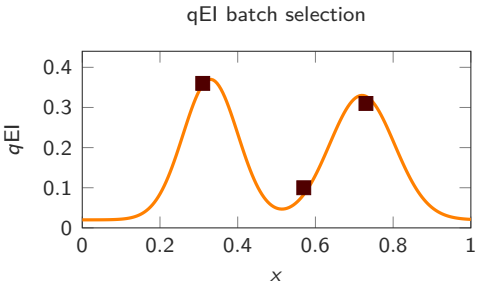
Instead of selecting one point $x$, we select a **batch** $X_q = \{x_1, \ldots, x_q\}$. For maximization,

$$\mathrm{qEI}(X_q) = \mathbb{E}\left[\max\left(\max_{i=1,\ldots,q} f(x_i) - y^+, 0\right)\right]$$

- qEI evaluates the **joint value** of a set of points
- it accounts for **posterior correlation**
- it avoids wasting a batch on nearly redundant experiments

# Why qEI is better than picking the top-$q$ EI points independently




**Greedy issue**

If we simply take the top few EI peaks, we may choose points that are **too close together** and provide overlapping information.

**qEI advantage**

qEI prefers a **diverse but valuable batch**: two points in different promising regions, and possibly one uncertainty-reducing point.

# Why batch acquisition functions matter in practice

## Benefits for lab workflows

- better use of parallel hardware and lab time
- lower wall-clock time to discover good candidates
- fewer redundant experiments in the same batch
- more robust progress when experiments are noisy

## Materials examples

- test 4 heat-treatment schedules at once
- fabricate several nearby alloy compositions together
- launch 8 simulations simultaneously on a cluster
- allocate one batch to both optimization and learning

## Take-home message

Sequential EI is great when experiments are strictly one-at-a-time. qEI is better when the real bottleneck is **wall-clock time** and experiments can be run in parallel.

# Conclusions

# Takeaways

- **Active learning** chooses experiments adaptively instead of blindly
- In materials design, this is valuable because data are expensive and scarce
- **Expected Improvement** balances:
  - exploitation of good predicted regions
  - exploration of uncertain regions
- EI can be dominated either by:
  - a very promising mean prediction
  - a highly uncertain region
- **qExpectedImprovement** extends this logic to batches and is useful for parallel experiments

# Thank You!

Questions?